Galaba Vamsi

Mechatronics at IIT Bhilai. Full-stack, AI/ML, and open-source robotics.
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SUMMARY

Graduating May 2027. 15 merged pull requests to open-source robotics tools: HFlow (robot-data conversion), Inspect Robots (evaluation), and Cerulion (runtime). I wrote and maintain inspect-robots-wandb on PyPI. Simulations in ROS 2/Gazebo, MuJoCo, MATLAB/Simulink, and Unity. Co-author of 2 RIS papers.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology Bhilai | **Coursework:** DSA, OOP, Deep Learning for Computer Vision, AI and ML, Digital Image Processing, Optimal Control, Advanced Control Theory. Aug 2023 – May 2027

OPEN SOURCE

HFlow Hebbian Robotics (YC S26) | Repository | My PRs Sep 2026 – Oct 2026

- Open-source SDK for verifying the quality of robot training data. 8 merged PRs, mostly making robot-log and LeRobot conversion keep data whole or refuse it with a clear reason, e.g. depth videos that would silently drop from 12-bit to 10-bit (#400).
- Benchmark PR (#367): timed three cold 1080p30 checks, reported no safe speedup, and left the filter graph unchanged.

Inspect Robots and inspect-robots-wandb Robocurve (YC S26) | Repository | PyPI | Plugin source Sep 2026 – Oct 2026

- 2 merged PRs: a configurable LLM retry policy across 5 wire protocols (#476) and unknown git state no longer logged as clean (#475). 3 open: #514 checkpoint/resume (open, in review), #550 on_eval_error sink hook, #551 plugin docs.
- Wrote and maintain inspect-robots-wandb, a community W&B plugin on PyPI, after the maintainers asked for the sink to live outside the core: 100% coverage gate, CI on Python 3.10–3.13, trusted publishing.

Cerulion Open-source Rust robot runtime that records a run and re-executes it against new code | Repository Oct 2026

- 5 merged PRs: 1 CLI fix (#216, ros2 attach --dry-run outside a workspace) and 4 documentation/comment corrections.

EXPERIENCE

Research Intern IIT Bhilai | Advisor: Dr. Arzad Alam Kherani | RIS-SIM v2 | v1 May 2025 – Jul 2025

- Built RIS-SIM, an open-source Python emulator for RIS-assisted radio links in LoS/NLoS scenarios with USRP-style SDR interfaces; designed RIS unit cells in CST Studio Suite for RF energy harvesting (IEEE INDICON 2025, co-author).
- Rebuilt in 2026 as RIS-SIM v2: six paper-grounded scenarios, a 20 Hz WebSocket dashboard, a ZeroMQ server, and 103 tests.

Co-founder, Human Slop Anti-AI social app | YC Startup School, Bengaluru | Web app | Play Store 2026 – Present

- Co-building it with my co-founder: an Expo (React Native) app for web and Android, a Fastify API, and PostgreSQL on Supabase, deployed on Render and Cloudflare. Live on the web and Google Play (50+ downloads).
- Added a device biometric check at signup and a keystroke-timing score in the editor; both are heuristics, not proof of authorship.

PROJECTS

Quarter-Car Active Suspension Study (course project, simulation only, 2026) Reproduced Na et al. (IEEE TSMC 2022) in MATLAB/Simulink: after fitting two plant parameters to its Case-10 experiment, the unchanged controller lands within about 6% of its error indices. The paper's Case-B gains fail a hold-out test, and LQR and skyhook match or beat it at equal effort. Site | GitHub

Whiteboard Cleaning Robot (2025, with Gajanand Kumawat) 3-DOF Cartesian gantry for a 90 × 90 cm whiteboard, simulated in ROS 2 Jazzy and Gazebo Harmonic: URDF/Xacro, ros2_control, 10 cm-overlap raster path, and logged coverage analysis. GitHub

Pneumatic Sorting Digital Twin (offline simulation) (2026) Unity simulation of three electropneumatic training kits: simulated axes, logical PLC I/O, HMI screens, and telemetry record and replay. Not yet connected to a physical PLC. GitHub

Charter (2026) Razorpay AI Buildathon prototype: chat, voice, and MCP (10 tools) share one bounded commerce core with frozen INR quotes, Razorpay test-mode payments, audit trails, and row-level security (TypeScript, Fastify, PostgreSQL). GitHub

Arista Wi-Fi RRM (team; Inter IIT Tech Meet 14.0, 5th place, Dec 2025) Client-aware radio resource management in NS-3 with I/Q-based interference classification (STFT, DBSCAN, Dual-CNN; 16 classes) on a 5-AP, 50-client 802.11k/v topology. Slides

AntennaNet (2026) Inverse antenna design: surrogate network plus KD-tree anchoring over about 2,000 simulated patch-antenna S11 spectra, extended to 12 × 12-pixel layouts checked by a method-of-moments solver; MEEP dataset scripts. GitHub

Lunar Crater Detection (2026) Crater extraction from lunar DEMs, reproducing Zhou et al. (2018): F1 0.839 on a 16-crater labelled region in Mare Serenitatis; an optional CuPy path makes preprocessing about 4× faster. GitHub

Prompt Robot Arm (2026) Kinematic MuJoCo simulation: a prompt such as "sort red, blue, green" is parsed (rule-based for now) into a pick-and-place plan, moving from a kinematic gripper to a SCARA arm with analytic inverse kinematics. GitHub

PUBLICATIONS

- **Co-author** — *Experience with RF Energy Harvesting-Driven Self-Powered RIS*, IEEE INDICON 2025.
- **Co-author** — *A systematic literature review on simulation models and deployments for reconfigurable intelligent surfaces*, Ad Hoc Networks (Elsevier), 2026.
- **Co-author (submitted, under review)** — *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, CoRL 2026 Workshop SPAIS.
- **First author (manuscript in preparation)** — sample-accurate IQ-level emulator for timed RIS control with controllers in the loop.

SKILLS

Robotics, simulation, and controls: ROS 2 (rclpy, ros2_control, URDF/Xacro), Gazebo Harmonic, MuJoCo, Unity (C#), MATLAB/Simulink, LQR and Kalman filter design, LeRobot datasets
Languages: Python, TypeScript/JavaScript, C/C++, Rust (open-source contributions), SQL, GLSL
ML, data, and scientific computing: PyTorch, PyTorch Geometric, OpenCV, NumPy, CuPy, Apache Arrow, Weights & Biases, MEEP (FDTD), CST Studio Suite, NS-3
Software, cloud, and tooling: FastAPI, Node.js (Fastify), React, React Native (Expo), Flutter, PostgreSQL, Supabase, WebSockets, Docker, AWS (CDK, Lambda, Step Functions, Bedrock), Git, Linux, GitHub Actions, pytest, Ruff, mypy, PyPI trusted publishing

ACHIEVEMENTS

AIR 200, Amazon ML Challenge 2026 (team) | **5th place**, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team, Dec 2025) | **AIR 700**, Amazon ML Challenge 2025 (team; 20,000+ participants) | **1st place**, IIT Bhilai Web Development Competition | **Top 10 finalist**, Toyota Hackathon | **Top 1%**, JEE Advanced | **PMSS** scholarship (2024)